

Syed Ali Hassan Jaffri

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Final-year Computer Science student at NUST focused on the security of LLM and RAG systems. Final-year project builds a defence layer that detects memory and corpus poisoning in agentic RAG pipelines; alongside it, has shipped ML-based security tools including a deployed AI web-application firewall and co-authored a paper under review at IEEE FIT 2026.

EDUCATION

National University of Sciences and Technology (NUST) — BS in Computer Science Sept 2023 – May 2027

- Achieved 7th position countrywide in the Secondary School Certificate (SSC) examination.

EXPERIENCE

Blockchain Engineer (Intern), Özyeğin University, Istanbul (Remote) Jun 2026 – Sep 2026

- Building the blockchain integration layer for a TÜBİTAK-funded vehicle-routing research project: five role-based Solidity smart contracts (Hardhat, Ethereum/Sepolia) and a Python backend (FastAPI, web3.py) that logs route plans on-chain and triggers re-optimization.

Cybersecurity Intern, NUST & QuantumRomics Oct 2025 – Mar 2026

- Analyzed IPFS-based decentralized storage for cryptographic weaknesses exposed by quantum algorithms and proposed a phased hybrid post-quantum migration roadmap, contributing to a co-authored paper under review at IEEE FIT 2026.

AI & Technology Intern, Agristreams, NSTP (Islamabad) Jun 2025 – Aug 2025

- Built Neurowall, an ML-based security layer that detects threats across multiple attack categories, with a real-time dashboard centralizing monitoring into one live interface. (Python, scikit-learn, Flask, React, Node.js)

Student Researcher — Video Content Extraction, NUST Oct 2025 – Dec 2025

- Built a video-retrieval pipeline (shot detection, keyframe sampling, visual + textual feature extraction) and trained a Siamese neural network for clip-to-clip similarity across 10,000+ media files. (Python)

Cybersecurity Research Associate, UMT, Lahore Jun 2024 – Aug 2024

- Ran security assessments across 5 web systems (SQLi, XSS, CSRF), delivering a prioritized recommendations report that closed gaps in detection coverage.

PROJECTS

Detecting Memory & Corpus Poisoning in Agentic RAG — Final Year Project

Building a security layer that continuously monitors an AI agent's long-term memory and retrieved documents to catch poisoned, manipulated, or inconsistent content before it reaches the model. Adapting research-backed defences (MEM-SAD, RAGuard, TrustRAG) into a deployable framework that hardens retrieval and audits memory without modifying the underlying LLM. (Python, LangChain/LlamaIndex, FAISS/Chroma, embeddings, Docker, REST APIs)

AegisVault — AI Web Application Firewall (live demo · code)

Real-time ML detection of malicious HTTP requests, with feature engineering across combined datasets to keep false positives low; deployed with an interactive dashboard that surfaces the layered decision pipeline and classifier confidence. (Python, scikit-learn, Flask, React)

AI-Driven Decentralized Traffic Routing Simulator (code)

Combines optimization (Dijkstra, GA, VNS), ML congestion prediction, and Ethereum smart contracts for tamper-proof on-chain auditing. (Python, FastAPI, Web3.py, Solidity, NetworkX)

Skin Lesion Classification (HAM10000) (notebook)

Deep-learning study classifying dermoscopic images across 7 diagnostic classes — from a CNN baseline up to a fine-tuned ResNet50 (transfer learning), with data augmentation and class-imbalance handling (weighted loss and sampling). (PyTorch, torchvision)

Credily (code)

Social platform with anonymous reviews, sentiment analysis, and a connection recommendation engine. (React, Node.js, Express, Cloud NLP)

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL, Solidity

ML & LLMs: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, LangChain, NumPy, Pandas

RAG & Vector Stores: Chroma, FAISS, Ollama, embeddings & semantic search, retrieval hardening

Security: adversarial ML, prompt-injection & RAG-poisoning defences, Burp Suite, Wireshark, Nmap

Web & Blockchain: React, Next.js, Node.js, Tailwind CSS; Hardhat, web3.py, Ethereum

Tools & Data: Git, Docker, Google Cloud Platform, Jupyter; MySQL, MongoDB, SQLite, GraphQL

CERTIFICATIONS & LEADERSHIP

- **Certifications:** ML & Deep Learning Specialization (Andrew Ng, Coursera); Quantum-Safe Cryptography (IBM); CTF challenges (CryptoHack).
- **Outreach Lead, ACM Student Chapter (NUST):** secured sponsorships and partnerships for technical events.